



Project-Based Learning (8 Industries We Tap)

- 1 **Customer Support**
AI assistants · ticket automation
- 2 **Sales & CRM**
Agentic outreach · pipeline updates
- 3 **Software Engineering**
Code agents · doc & test generation
- 4 **Operations & PMO**
Workflow automation · reporting
- 5 **Finance & Analytics**
Document extraction · ETL agents
- 6 **Healthcare**
Clinical Q&A · summarization
- 7 **Legal & Compliance**
Contract analysis · RAG search
- 8 **Marketing & Content**
Content generation · multi-agent

CHAPTER 1

GenAI Foundations & LLMs

From Python & ML basics to how large language models actually work

TOOLS

Python · Pandas · NumPy · Hugging Face · OpenAI · Gemini

TOPICS

- AI · ML · DL · NLP
- TF-IDF & Word2Vec
- Tuning & Adaptation
- Context Window & Tokens
- Knowledge Cut-off
- Guardrails
- BoW & One-Hot
- Transformer Architecture
- Open vs Closed Source
- Decoding Parameters
- Hallucination
- Human in the Loop

PROJECTS

- LLM Playground & Token Explorer
- Decoding Parameter Lab

→ Understand how LLMs generate text → Build intuition for tokens & sampling

CHAPTER 2

Prompt Engineering & LLM APIs

Designing reliable prompts and structured outputs around model APIs

TOOLS

OpenAI · Anthropic · Gemini · Pydantic

TOPICS

- LLM API Integration
- Few-Shot & CoT
- Output Parsers
- Prompt Injection
- Prompt Types
- Structured Outputs
- Schema Enforcement
- Prompt Evaluation

PROJECTS

- Structured Output Extractor
- Prompt Injection Defense System

→ Get reliable, typed outputs from LLMs → Defend prompts against injection

CHAPTER 3

RAG Pipelines

Grounding LLMs in your own data with retrieval pipelines

TOOLS

LangChain · pgvector · Qdrant · Cohere · Jina

TOPICS

- Loaders & Splitters
- Embeddings
- Retrieval Config
- Advanced RAG
- Chunking Strategies
- Vector Databases
- Reranking
- RAG Evaluation

PROJECTS

- Document Q&A RAG System
- Advanced RAG with Reranking

→ Ground answers in private data → Tune retrieval for accuracy

CHAPTER 4

Agents & Agentic Patterns

Turning static Q&A into agents that reason, plan and call tools

TOOLS

LangChain · LangGraph · CrewAI · Agno · AutoGen

TOPICS

- Agent Architecture
- The Agentic Loop
- Agent Memory & Types
- Multi-Agent Systems
- Agentic Patterns
- Tools & Calling
- Agent Conversation
- Supervisor / Worker

PROJECTS

- Tool-Calling Research Agent
- Multi-Agent Content Crew

→ Build agents that use tools → Coordinate role-based agent teams

CHAPTER 5

Orchestration, Frameworks & MCP

Production orchestration with the LangChain stack, MCP and no-code

TOOLS

LCEL · LangGraph · LangSmith · MCP · n8n · Langflow

TOPICS

- LCEL & Templates
- Evaluator-Optimizer
- Bedrock & SageMaker
- MCP Tools
- Routing & Parallelism
- LangSmith Tracing
- MCP: Host · Server
- n8n & Langflow

PROJECTS

- LangGraph Workflow Orchestrator
- MCP-Powered Desktop Assistant
- n8n Business Automation Pipeline

→ Orchestrate multi-step agent workflows → Connect agents to real systems

CHAPTER 6

Safety, Eval, Serving & Deploy

Guardrails, evaluation and shipping agents to production

TOOLS

NeMo Guardrails · Ragas · TruLens · Ollama · vLLM · Docker

TOPICS

- Safety & Guardrails
- Ragas & TruLens
- Ollama & vLLM
- Docker & Cloud
- NeMo Guardrails
- Phoenix Monitoring
- LangServe Streaming
- AWS · GCP · Azure

PROJECTS

- Guardrailed, Evaluated Chatbot
- End-to-End AI Copilot

→ Make agents safe & measurable → Deploy a working AI Copilot

GenAI foundations are a must before Autonomous Agents.